

- 1) Business analytics - uses data + stats to solve problems and drive strategy, bigger picture than just
2. Data analytics - the process of examining raw data patterns, more technical/hands-on than business analytics.

Difference: business ~~analytics~~ analytics = business - for data analytics = process-focused. BA uses DA.

3. Business intelligence (BI) - Tools/dashboards show what's happening now. Think power BI, Tableau.

VS BA: BI = reporting past/present. BA = includes future

4. Data-Driven Decision making - Philosophy: use facts not gut feel, to make choices.

The Data itself

5. Big Data - Data too big / Fast / Varied for normal tools. 3 V's. Volume, Velocity, Variety.

The 4 types of analytics - (The analytics)

6. Descriptive analytics - what happened? Summaries, totals. Example: sales dropped 10% last month.

~~2 Predictive Analytics~~ Diagnostic analytics

~~What did it happen?~~ Forecasts, ML models, etc sales w

7. Diagnostic analytics - why did it ^{happen} " ? " drill down, correlate
example: sales dropped because of stock-outs.

8. predictive analytics - what is likely to happen? " for
ml models. example: sales will drop 5% next month.

9) prescriptive analytics - "What should we do?" recommen-
dations, simulations. example: order more stock now to prevent

10. Data literacy - ability to read, understand, question
and argue with data, key skill for everyone, not just ana-

11. structured data - neatly organized in rows/columns. easy
computers to read: example: excel spreadsheet, sql database

12. Unstructured data: - no predefined format, mess, email
emails.

13. Data warehouse - central library where clean, structured
data from any source is stored for analysis. example: sales

14. Data mining - digging through large ~~datasets~~ datasets
find hidden patterns, trends, or relationships. example: find
that customers who buy milk often buy beer too.

15. Data visualization - turning data into charts, graphs,
makes patterns easier to see. example: pie chart of sales by product

16. Dashboard - live screen showing key charts/metrics in
place. updates automatically. example: stock levels

17. metric - Any number you can measure. Need
example website visits

18) Key performance indicator (KPI) - A metric that matters business goals, drives decisions. Example: "Monthly sales KPI." "Number of staplers in office" is just a metric.

19) Benchmark - A standard or point you compare.
example: Industry average delivery time = 2 days. You = 3 days = below benchmark.

20. Segmentation - Splitting data into groups with similar characteristics.
example: Grouping customers by age, location.

21. Conversion rate - % of people who do what you want.
Formula: $\text{Conversions} \div \text{Visitors} \times 100$

23. Customer lifetime value (CLV/LTV) - Total profit a customer brings before they leave. Formula: $\text{avg order value} \times \text{avg years}$.

22. Customer acquisition cost (CAC) - Total cost to acquire a new customer. Formula: $\text{Sales \& marketing spend} \div \text{New customers}$.

24) Churn Rate - % of customers who stop buying/leave.

25. Customer retention rate - % of customers you keep. Opposite of churn.

26. Net promoter score (NPS) - loyalty measure. Question: "Would you recommend us?" 0-10 scale.

27. Average order value (AOV) - average revenue per transaction. Formula: $\text{total revenue} \div \text{number of orders}$.
example: \$5000 sales from 250 orders $\rightarrow$ AOV = \$20.
business question: How do we get customers to spend more per visit?

28. Sales Funnel - stages a customer goes through: Awareness
interest $\rightarrow$ decision $\rightarrow$ purchase. Example: 10000 see ad $\rightarrow$ 1000 click
 $\rightarrow$ 100 add to cart $\rightarrow$ 20 buy. business question: are we losing customers?

29. ~~marketing~~ marketing campaign - coordinated set
activities with goal, budget, and timeframe. Example
~~Capitol~~ Capital's "we butter" campaign across TV, radio.

30. Post-campaign analysis - Reviewing results after
campaign ends to see what worked.

uses: ROI

31. Incremental Revenue - extra sales caused only
the campaign. Not total sales

32. Return on investment (ROI) overall profitability of
campaign. includes all costs. Formula: $\frac{\text{Revenue} - \text{Total cost}}{\text{Total cost}} \times 100$

33. Return on ad spend (ROAS) - Revenue per \$1 of
ad spend only. ignores staff/agency costs

34. Click Through Rate (CTR) - % of people who saw
ad/link and clicked it. Formula: $\frac{\text{clicks}}{\text{impressions}}$

35. Bounce Rate - % who land on your site and
without any action.

36. A/B Testing - compare 2 versions to see
performs better.

37. Lead generation - capturing contact details of potential customers.

38. Marketing attribution - Figuring out which gets credit for a sale.